MAGNETIC LEVITATION IN TRAINS

Introduction: The modernisation of society has continually reshaped the way people live, work, and travel. As populations grew and cities expanded, the use of traditional transportation often lacked the requirements of the fast paced and industrialised society. With rail transport being one of our most revolutionised modes of transport, it faced many challenges, such as noise pollution, energy inefficiency, and mechanical limitations caused by friction between wheels and rails, which has caused us to turn to Magnetic Levitation Systems in trains.

Concept of Magnetic Levitation

1. Magnetic Levitation (Maglev) eliminates friction through basic magnetic rules where poles repel.
2. Propulsion done by electromagnetic coils that switch on and off to alternate polarity.
3. Propulsion can also be done through alternating polarity where repulsion pushes the train whilst attraction pulls the train; magnets arranged in Halbach array.

Types of Magnetic Levitation Systems

1. <u>Electromagnetic Suspension (EMS):</u> Electromagnets on the train bottom attract oppositely charged magnets on the steel rails, gap between train and track is constantly maintained by changing the electromagnetic force according to feedback by sensors.
2. <u>Electrodynamic Suspension (EDS):</u> Superconducting magnets on the train pass over coils on the track, creating eddy currents which create repulsive forces with the train's magnets, keeping it levitated at high speeds.

Benefits of Magnetic Levitation (as compared to traditional trains)

Due to the lack of friction, magnetic levitation makes trains faster, quieter and smoother, while vastly reducing mechanical wear and tear.

Real Life Applications of Magnetic Levitation Systems

China: Shanghai Maglev Train (EMS)

The Shanghai Maglev Train is the world's first high-speed maglev system which began their commercial operation in 2004. This fast-paced train operates at speeds up to 431 km/h, allowing people to complete their journey in less than 8 minutes.

Japan: Chuo Shinkansen (EDS)

The Shinkansen Maglev Trains are amongst the fastest in the world, operating at speed of up to over 500 km/h, vastly reducing travel times. The system is slated to fully open in 2036 for commercial use.

Conclusion

Magnetic levitation technology represents a major advancement in modern transportation by overcoming the limitations of traditional rail systems. By removing wheel rail contaect, magnetic levitation trains significantly reduce friction which results in higher speeds, smoother rides and reduced mechanical wear. Systems such as EMS and EDS demonstrate how magnetic forces can be effectively used to levitate and propel trains with precision and efficiency. As such, magnetic levitation offers a promising solution for fast, efficient, and sustainable transport in an increasingly civilized and industrialised world.